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Automated Percutaneous Lumbar Discectomy (APLD), Low Back Pain

ACG: A-0396 (AC)

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Clinical Indications for Procedure

- Current Role Remains Uncertain. Based on review of existing evidence, there are currently no clinical indications for this technology. See the Inconclusive or Non-Supportive Evidence section for more detailed analysis of the evidence base.(1)(2)(3)

Alternatives to Procedure

- Alternatives include(10)(11):
 - Lumbar discectomy or microdiscectomy
 - Nonoperative measures, including(6):
 - Epidural corticosteroid injection. See Epidural Corticosteroid Injection [AC](#) for further information.
 - Patient education in self-management and exercise
 - Pharmacotherapy (eg, anti-inflammatory medications)(12)(13)(14)
 - Physical therapy. See Spine Soft Tissue Dysfunction Rehabilitation [AC](#) for further information.
 - Spinal manipulation therapy. See Spinal Manipulation Therapy (SMT), Chiropractic and Other [AC](#) for further information.

Evidence Summary

Background

Automated percutaneous lumbar discectomy is a minimally invasive technique to reduce the pressure of a small contained herniated lumbar disk by mechanically cutting and aspirating portions of the nucleus using a probe inserted percutaneously under fluoroscopic guidance.(4)(5)(6) (**EG 2**)

Inconclusive or Non-Supportive Evidence

The evidence for the recommendations found in this guideline includes 6 published peer reviewed articles, 1 Cochrane systematic review, and 2 book sections.

For lumbar disk disease, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. (**RG B**) A systematic review of limited data from small randomized controlled studies with

insufficient long-term follow-up concluded that automated percutaneous lumbar discectomy resulted in inferior results compared with other, more established procedures, including microdiscectomy; the review also concluded that the procedure should be used as a research technique only.(1) **(EG 1)** A network meta-analysis of 29 randomized controlled trials evaluating the efficacy of 7 surgical interventions for lumbar disk herniation (including percutaneous endoscopic microdiscectomy, standard open discectomy, and standard microdiscectomy) found that, compared with other surgical interventions, automated percutaneous lumbar discectomy was associated with the worst outcomes for increasing surgical success, decreasing complication rates, and decreasing reoperation rates. Higher-quality randomized controlled trials with direct comparisons between the surgical alternatives were recommended.(2) **(EG 1)** A systematic review of 19 observational studies (a total of 5515 patients) with a minimum of 1-year follow-up that evaluated the efficacy of automated percutaneous lumbar discectomy found that the evidence was limited for short-term and long-term relief. However, the authors noted that the procedure may provide pain relief for selected patients with contained lumbar disk herniation.(3) **(EG 1)** An observational study of 147 patients with clinical and radiologic diagnosis of sciatica found that automated percutaneous lumbar discectomy was associated with improved pain and symptom scores as compared with caudal epidural at 6-month follow-up. Additional studies with larger patient groups and longer follow-up were recommended.(7) **(EG 2)** A study of 100 patients reported, at 6-month follow-up, that 80% of patients with lateralized herniated disks achieved at least a 70% reduction in pain scores, as compared with 50% of patients with a posteromedian herniated disk. Of the entire cohort, 70% of patients ceased use of analgesics, and 10 patients required repeat surgery.(8) **(EG 2)** For recurrent lumbar disk herniation, a study of 6 patients reported that, although 4 patients had significant improvement in sciatica pain scores at follow-up ranging from 5 to 8 months, 2 patients had no improvement and required surgery within 3 weeks of the procedure.(9) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment across patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 422.101(15); 42 CFR 419.22(16)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(17); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(18); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(19)

Medicare Coverage Determinations: Medicare Coverage Database(20)

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